

SCIENCE FOR POLICY BRIEF

Green Patenting of EU vs. Global Competitors in the Digital Techno Economic Ecosystem

2025

HIGHLIGHTS

- ▶ Although EU players (firms, research institutions, and government bodies) have increased their twin technology (green and digital) patents by an average of 14% annually since 2009, the EU faces challenges in enhancing its competitive stance against global players, particularly China.
- ▶ It is crucial to monitor the evolution of the EU's international partnerships regarding co-application in twin patents to avoid becoming overly dependent on foreign collaborations.
- ▶ Most EU firms engaged in twin transition innovation are mainly under domestic control, reflecting the global trend.
- ▶ The EU reached by 2022 a competitive edge in twin technologies related to building and reducing greenhouse gas emissions in energy processes, enhancing the good position registered in 2019.
- ▶ Beyond the digital ecosystem, energy-intensive industries and electronics were the leading ecosystems in EU patent-driven twin innovation from 2009 to 2023.

Quick Guide

This analysis, along with the brief 'Green and digital patenting of Industrial R&D Investment Scoreboard firms' (Ince and Georgakaki, 2025), examines patent-driven innovation for twin transition through two distinct but complementary approaches, with a particular focus on Scoreboard firms. Our approach uses 'green' patent codes to explore twin innovation within the Digital Techno-Economic ecoSystem (DGTES) to analyse EU's global position and its industrial composition. In contrast, Ince and Georgakaki (2025) investigate how the intersection of climate change mitigation technologies and digital technologies either transforms our interaction with energy systems or mitigates their additional energy consumption. Despite differences in methodology, both studies converge on key messages, strengthening the validity of the findings. These shared conclusions are cross-referenced in both documents.

Introduction

The ongoing twin transition—simultaneously pursuing the digital and green transitions—presents both a challenge and an opportunity to move closer to the needed level of transformation for an environmentally sustainable and competitive European Union (EU). The relationship between the two transformations is complex. Rebound effects from digitalization such as energy consumption and e-waste can hinder the green transition (Muñoz-de-Bustillo Llorente, 2024). However, digital technologies are pivotal in tackling climate change through innovation, accelerating the EU's competitive edge and productivity.

Box 1: DGTES methodology

We identify the innovation activities that support the twin transition to a green and digital economy. The approach's granularity allows for examining innovation's evolution, main features for the twin transition and benchmarking EU's position relative to global competitors such as China and the US.

The basis of our analysis is 2009–2023 global patent filings sourced from Patstat, which contain textual information on the invention description and the applicant. First, we search a comprehensive dictionary of keywords on digital technologies in the patent descriptions. To get the subset of twin innovation activities, we filter for patents on climate change mitigation technologies (Corporate Patent Classification (CPC) code Y02).

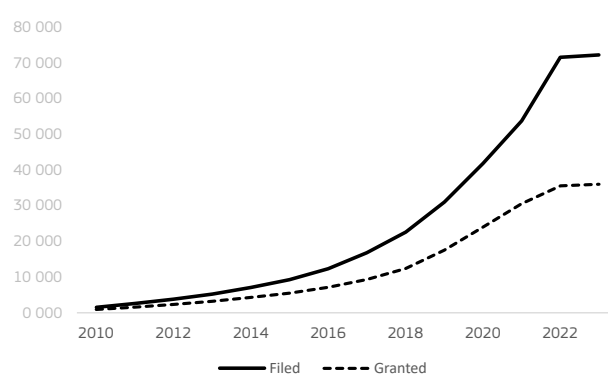
We use information on patent attributes to identify who is innovating in the twin transition (the players), their geographical distribution, and the connections among them as they co-patent, forming a network of collaborations within and across borders. We complement the data with information on (i) the industrial ecosystems associated with each player (Signorelli et al., 2025), and (ii) the firm's historical ownership from Orbis. We exploit this network and generate a series of indicators to analyse the EU's competitiveness and foreign dependencies as regards the twin transition.

This study explores how the green and digital transitions reinforce each other, pushing the innovation frontier. We look at the intersection of these two transformative processes by leveraging patent data and text analysis to identify emerging climate technologies at the heart of the twin transition. We rely on the Digital Techno-Economic ecoSystem (DGTES) methodology from Calza et al. (2022) to identify priority patent applications for technologies related to the twin transition. By constructing a network of organizations engaged in co-application, this analysis offers new insights into the state of patent-driven innovation on twin technologies in the EU, its interlinkages with other global regions, and its prevalence across industrial ecosystems.

Twin transition in the EU from a global perspective

This section analyses the EU's position in the global patent-driven innovation landscape related to the twin transition. It highlights the findings of the mapping exercise that examined the geographical distribution of players filing priority patents in digital technologies, covering those aimed at climate mitigation (referred to as twin patents).

Figure 1 – Global patents filed vs. granted on digital and climate mitigating technologies, 2009–2023.



Note: Cumulative sum of patents worldwide filed on digital technologies that also include climate-mitigating technologies (CPC code Y02). *Source:* DGTES database.

Patents on the twin transition grew exponentially over the last decade, cumulating over 36 000 granted patents globally by 2023. The number doubles when accounting for filed patents i.e., including non-granted ones (**Figure 1**). Filed patents indicate emerging innovation trends and competitive activity, even before being granted. They provide valuable insights into areas of technological focus and R&D investment, which may signal future

advancements, even if not all patents are ultimately granted.¹

Figure 2 – Digital patents filed on climate-mitigating technologies by geographical area, 2009–2023.



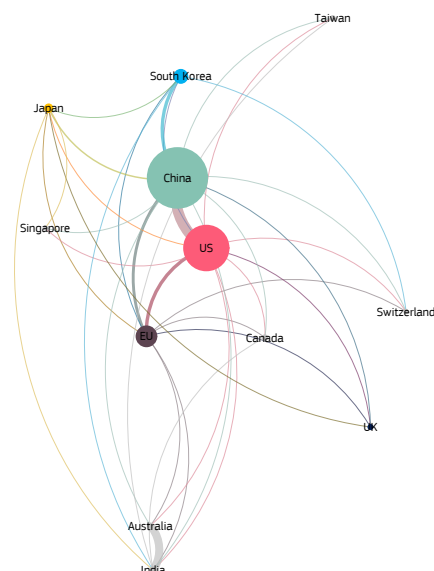
Note: Patents filed by geographical area on digital technologies that also include climate-mitigating technologies (CPC code Y02). Collaborative patents count on a fractional basis, i.e. divided by the number of co-applicants of the patent. *RoW* stands for rest of the world. *Source:* DGTES database.

Even though the number of twin patents filed by EU players has increased by an average of 14% annually since 2009, the analysis underscores the ongoing challenge for the EU to strengthen its position relative to other global players, especially China. China holds the largest share of twin patents (79.4%), surpassing the US (6%) and the EU (2.4%) (**Figure 2**). This finding holds even when considering only granted patents, and highlights China's dominant role in driving innovation in digital and climate-mitigating technologies.²

The leading role of China in the global scene is further supported by the network of global collaborations on twin patents based on international co-applications (**Figure 3**). China has a central role as it presents the highest betweenness centrality score (=1), serving as a crucial link or bottleneck between regions.³ The US holds a strategic position in the global flow of knowledge (score 0.87), exerting high influence in the network and having the highest number of co-applications with Chinese entities in the network. The EU's influence in the global network of patent-driven

innovation of twin technologies is limited (score 0.29) and its tightest partnerships are with US entities, followed by China, India, Canada, and Switzerland.⁴

Figure 3 – Global collaborations in patent-driven innovation on digital and climate mitigating technologies.



Note: The network represents international co-applications in digital patents that cover climate mitigating technologies filed between 2009 and 2023. The nodes group players by country (the node EU refers to EU27) and their size depends on the total number of international collaborations. The edges represent the bilateral collaborations between the nodes. *Source:* DGTES database.

Next, the focus turns to the linkages of players with other entities to help identify the strengths and dependencies of the EU twin innovation landscape. In this setting, two organizations can connect through two types of relationships: co-applicants of a patent and firm ownership, signalling control over the innovation.

First, the analysis of patent collaborations reveals a relatively balanced relationship with the US but an imbalanced relationship with China. China and the United States are the primary collaborators for EU organizations, with 6.9% of EU co-applications involving Chinese partners and 8.2% involving US-based partners (**Table 1**). However, there is a significant imbalance in the relationship with China: only

¹ See Pasimeni and Georgakaki (2020) for main concepts for patent analysis and data availability time lags.

² The finding on Chinese global leadership on twin patents is also robust to counting patents filed up to 2020, though with a smaller difference across regions (CN: 69%; US: 9%; EU: 4% of global patent share). This finding aligns with the relatively low and stagnated patenting activity of the EU versus China documented by Ince and Georgakaki (2025) using high-value green inventions between 2016 and 2020.

³ Betweenness centrality is a network indicator that measures the relative position of a node (country in this case) and the extent to which a node serves as a key broker between many other nodes. It is helpful for identifying key players or influential nodes within a network as it computes how often a node lies in the shortest path between two nodes in the network. The value is normalised between 0 and 1, where a country with a high intermediation score in the innovation network is considered a gatekeeper of knowledge flowing across global regions.

⁴ To assess partnerships' tightness (the width of the edges), we consider all collaborations between each node-pair in the network. The value is normalised between 0 and 1, where the edge with the highest number of collaborations is US-China. EU collaborations with US-based partners get a value of 0.32, while with Chinese partners 0.27.

0.4% of China's collaborations involve EU co-applicants. Instead, the EU exhibits a more balanced co-operation with the US, accounting for 6.1% of US co-applications. It is crucial for the EU to monitor the evolution of its collaborations with China carefully to avoid becoming overly dependent on Chinese partnerships, which could limit its strategic autonomy in the long-run.

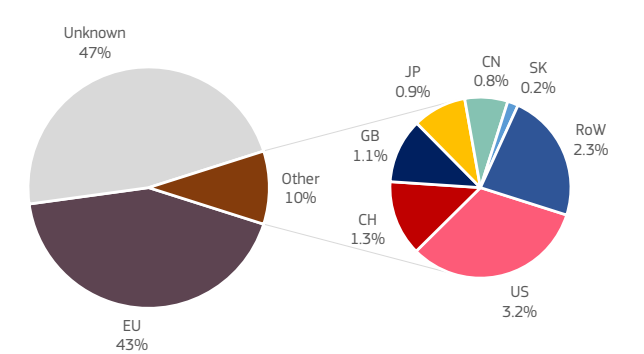
Table 1 - Affinity matrix: dependencies in green and digital patents (2009-2023).

		...that are collaborative with					
		CN	US	EU	SK	JP	RoW
Share of patents filed by	CN	97%	1.4%	0.4%	0.4%	0.2%	0.4%
	US	19.1%	66%	6.1%	0.5%	0.5%	7.5%
	EU	6.9%	8.2%	78%	0.0%	0.3%	6.5%
	SK	8.3%	0.7%	0.1%	90%	0.2%	0.4%
	JP	6.6%	1.2%	0.6%	0.4%	90%	1.0%
	RoW	3.3%	4.4%	2.8%	0.2%	0.2%	89%

Note: Patents filed by geographical area on digital technologies that also include climate-mitigating technologies (CPC code Y02). Cells indicate the share of collaborations between node in row *i* and node in column *j* over the sum of collaborations of node *i*. In the diagonal, i.e., when *i=j*, the cells display the shares of collaborative activities within the same geographical areas (internal collaborations). Row values add up to one. High asymmetry reveals an imbalance. RoW stands for rest of the world. Source: DGTES database.

Second, the analysis of ownership links indicates that EU firms involved in twin transition innovation are predominantly under domestic control, aligning with the global average. The dataset allows for tracking the location of the global ultimate owner for more than half of EU firms that filed at least one twin patent from 2009 to 2023.⁵ Out of the subset of EU firms with ownership information, 80% account as domestic—i.e., their ultimate owner is in the EU. This fact is further supported by the evidence in Ince and Georgakaki (2025) where domestic ownership is observed at the EU member state level. The remaining 20% of firms involved in twin patents have an ultimate owner outside the EU (**Figure 4**). The US is the primary partner, followed by Switzerland and the UK.⁶ This preliminary analysis highlights the need for further efforts to track ownership structures, as understanding these links can help inform policies that strengthen the EU's autonomy in emerging technologies.

Figure 4 – Ownership of EU firms with patents in both digital and green technologies as of 2023.



Note: Figures based on the tracked 2023 ownership status of the 531 EU firms identified as filing at least for one green and digital patent during 2009-2023. Foreign ownership defined as the global ultimate owner holding more than 50% of shares. A firm located in the EU is foreign when the owner is located outside the EU. Unknown refers to firms with no ownership data. RoW stands for rest of the world. Source: DGTES database.

Technologies behind the green transition

Breaking down the twin transition-related technologies helps assessing EU's competitiveness in each area compared to other global regions and regarding changes in the post-pandemic era.

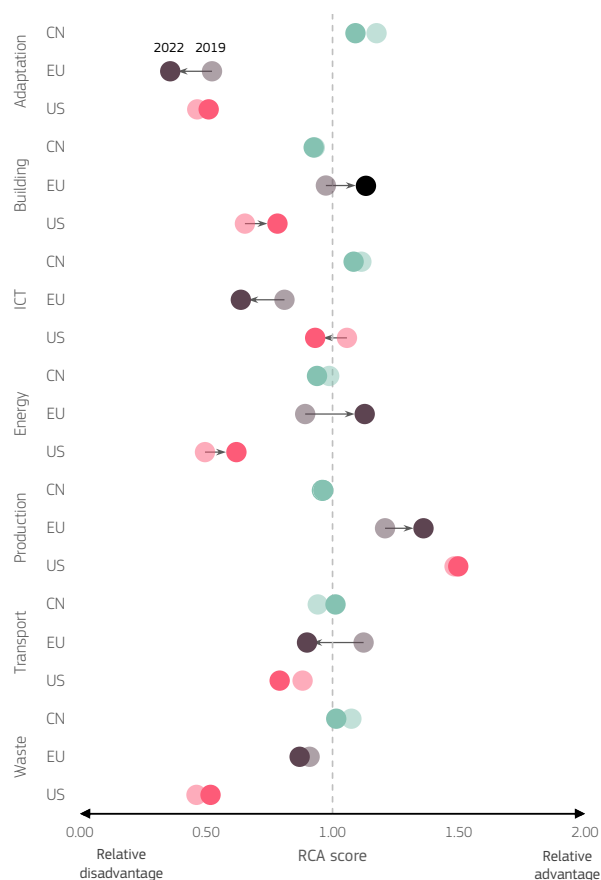
The relative comparative advantage (RCA) score measures a region's involvement in patent-driven innovation for a specific technology, compared to the global average. An RCA value greater than one implies that the geographical area has a higher-than-global average proportion of patents filed in that technology, suggesting the presence of a competitive edge in that technology. This analysis focuses on climate-mitigating technologies (CPC subclasses of Y02) and compares the EU's RCA score with those of the US and China in the years 2019 and 2022 as shown in **Figure 5**.

China leads in the number of twin patents across climate-mitigating technology subclasses. Still, it holds a relative competitive advantage over the EU and the US in only four areas: adaptation to climate change technologies (Y02A), information and communication technologies (or ICT, Y02D), transportation (Y02T), and waste management (Y02W).

⁵ The ownership analysis excludes academic institutions and government bodies filing for twin patents. The global ultimate owner is the entity controlling over 50% of the shares of a company. Foreign-owned firms have a global ultimate owner located in another country. A firm is foreign in the EU when the owner is located outside the EU, while ownerships between Member States are domestic. With half of the firms with known ownership, the EU is the region with the second highest share of tracked ownership after Japan (61%) (this figure is 45% for China, 35% for the US, and 11% for South Korea).

⁶ As shares of all EU firms filing for patents, 43% are domestic, while 10% are foreign. Unfortunately, ownership data for the remaining firms is unavailable.

Figure 5 – Relative comparative advantage of the EU, China, and the US in climate mitigating technologies patents: 2019 to 2022 change.



Note: Relative comparative advantage (RCA) of geographical area on patents filed on digital technologies that also include climate-mitigating technologies (CPC code Y02) by technology class. RCA score calculated by comparing the geographical area's share of patents filed in a specific technology in its twin patent portfolio against the global share in that same technology. RCA value greater than one means that the geographical area has a higher-than-world average proportion of patents filed in that technology. Light shaded bubbles represent 2019 values for patents filed during 2009-2019 while dark shaded bubbles represent 2022 values for patents filed during 2009-2022. *Source:* DGTES database.

The US maintains a strong advantage in climate change mitigating technologies in production processes (Y02P), consistent since 2019. Meanwhile, the EU has a competitive edge in technologies related to buildings (Y02B), reduction of greenhouse gas emissions related to energy processes (Y02E), and, following the US, in production technologies (Y02P). It holds the highest RCA in building and energy globally, strengthening its innovation in these areas since 2019. This innovative progress has allowed the EU to reach a competitive edge by 2022. Technologies reducing energy emissions register the largest improvement in the EU's RCA score between 2019 and 2022. However, from 2019 to 2022, the EU's share has declined relative to the global average in ICT, adaptation, and transport technologies. The latter group shows the largest

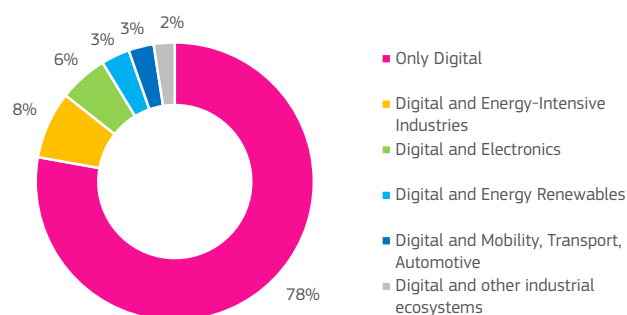
drop in RCA score across all technology classes. While it already faced disadvantages in ICT and adaptation in 2019, these worsened, and EU's RCA in transport has shifted from a competitive edge to a disadvantage.

Twin transition in the EU across industrial ecosystems

This section expands the DGTES methodology to other industrial ecosystems beyond digital and explores how twin innovation is transversal to the EU's industries.

All patents are part of the digital ecosystem, which is central to the twin transition; nearly 80% of patents filed by EU players are not directly connected to other industries and focus solely on digital technologies (**Figure 6**). Nevertheless, some of these technologies can still indirectly provide spillover efficiency gains to other industrial ecosystems. For example, general-purpose technologies like communication networks can lead to productivity gains or process improvements across multiple industries.

Figure 6 – Engagement of industrial ecosystems in the EU on green and digital innovation.



Note: Patents filed by EU players on digital technologies that also include climate-mitigating technologies (CPC code Y02) during 2009-2023. Collaborative patents count on a fractional basis, i.e. divided by the number of co-applicants of the patent. Figures aggregated at the EU level. Patents corresponding to multiple ecosystems get a weight based on the number of overlapping ecosystems. *Source:* DGTES database.

Energy-intensive industries and electronics were the most active ecosystems in filing digital and green innovations patents between 2009 and 2023. Together, these industries account for 14% of all twin patents filed by EU players. For instance, the electronics sector has been key in developing energy-efficient devices. At the same time, energy-intensive industries focus on reducing carbon emissions through digital solutions like smart sensors and AI for energy optimization.

Box 2: Twin transition in Scoreboard firms

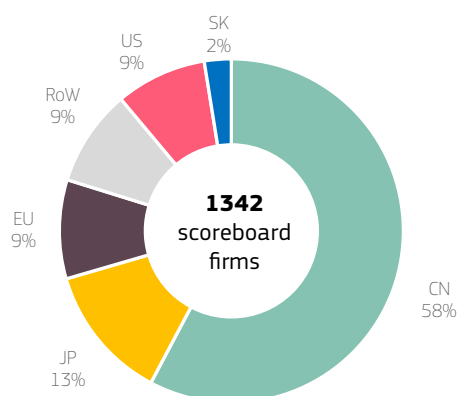
'The 2024 EU Industrial R&D Investment Scoreboard' includes 3% of companies filing for twin patents in this analysis. This box examines the distribution of these firms across technologies and countries.

China leads with 58% of firms filing for twin patents, reflecting trends similar to non-scoreboard firms. Japan is next with 13% of these firms, while the EU and the US each host 9% (**Figure 7**). Scoreboard firms primarily innovate in climate mitigation technologies related to production processes (Y02P), with over half holding patents in this area. They also focus on ICT (Y02D, 30%) and energy (Y02E, 29%) technologies for the twin transition (**Figure 8**).

Doubling the share for non-scoreboard firms, nearly 40% of scoreboard firms seek patents in multiple technology areas, with 22% active in two climate-mitigating technologies and 18% engaged in three or more.

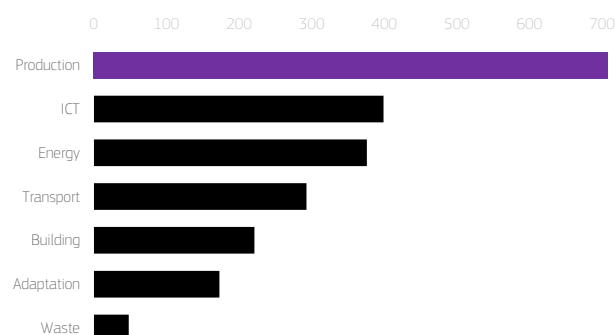
Among these firms, General Electric (US), stands out as the firm holding the largest number of twin inventions. It accounts for over 400 twin patent filings between 2009 and 2023. Chinese Tencent (technology), Naura (semiconductor), and Huawei (telecommunications) complete the podium with over 100 twin patents filed each during the period.

Figure 7 – Share of Scoreboard firms filing for twin patents by geographical area.



Note: Geographical distribution of scoreboard firms with patents filed on digital technologies that also include climate-mitigating technologies (CPC code Y02). *RoW* stands for rest of the world. *Source:* DGTES database.

Figure 8 – Number of Scoreboard firms filing for twin patents by climate mitigating technology.



Note: Number of Scoreboard firms with patents filed on digital technologies that also include climate-mitigating technologies (CPC code Y02) by technology class. *Source:* DGTES database.

Conclusions

This study underscores the EU's progress and challenges in the twin transition, offering insights into the EU's competitive position and potential areas for policy action. Despite twin patents filed by EU players increasing on average of 14% annually since 2009, the number of patents is still far behind those from China and the US, clear leaders in twin technologies innovations. China and the US host also most co-applicants for patent filings of EU-based players, and the study underscores potential dependencies on Chinese partnerships.

The findings highlight the importance of fostering patent-driven innovation, so the EU can set a global benchmark for integrating digital and green solutions. The analysis outlines that the EU reached a competitive edge in twin technologies related to building and reducing greenhouse gas emissions in energy processes while is lagging its competitors in ICT, adaptation, and transport technologies.

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